Ahmad Alqaisi

Indiana University

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Education

Ph.D., Physics (Experimental High Energy Physics), Indiana University, US, August 2024 – Present
Advisors: Jon Urheim
M.S., Materials Science, and Engineering, Arizona State University, US, May 2023
Thesis title: A Computational Study on the Melting Point of Si-Ge-Sn high-entropy alloy
Advisors: Qi-Jun Hong, Houlong Zhuang
GPA: 3.67/4.0 (Magna Cum Laude)
B.Sc., Physics, Al-Balqa Applied University, Jordan, February 2020
Thesis title: The Physics Principles of Super Resolution Microscopy
Advisors: Mazen Nairat
GPA: 3.33/4.0 (Very good)

Research Interests

High Energy Physics, Neutrino Physics, Liquid Argon Time Projection Chamber (LArTPC), Photon Detection System (PDS)

Technical Skills

- LArSoft
- ROOT/PyROOT
- C++/Python
- High-Performance Computing, Linux
- Machine Learning (TensorFlow, PyTorch)

Professional Experience

Neutrino Physics, Indiana University Neutrino Group (IUNeutrinos), March 2025 – Present
Working with Professor Jon Urheim within the DUNE Experiment Collaboration. The main theme is Photon Detection System (PDS):

- Studying Rayliegh Scattering Length (RSL) using simulations in LArSoft code in the ProtoDUNE-VD detector

- Generating simulation data via LArSoft and training the Machine Learning (Computable Graph Module) Fast Optical Simulation model with TensorFlow package for the ProtoDUNE-VD detector for different RSL values

Associate Instructor, Indiana University Bloomington, Department of Physics
January 2026 – May 2026
Instructing six discussion sessions per week for the course: Introductory Electricity and Magnetism with Calculus **(P222)**. Other duties included executing exam review sessions and exam grading for the students.

August 2024 – May 2025, August 2025 – December 2025
Instructing two sessions of introductory Physics labs **(P201)** per week. Other duties include preparing short lectures containing information needed for the lab, guiding students with lab procedures and grading lab reports.

Center for Exploration of Energy and Matter, Indiana University Bloomington, May 2023 – Fall 2024
Experimental Nuclear Physics training at the IUB *Center for Exploration of Energy and Matter (CEEM)* under the supervision of Professor Michael Snow. The training and research activities included the following:

- **NNBAR Collaboration (Fall 2023 – Fall 2024):** Training in Neutron Optics theory and Neutron ray tracing using the McStas code. Analysis of Neutron reflection probabilities of Neutron guides from previous NNBAR experiments based on novel Guide Reflectivity models using codes such as McStas, VITESS and GEANT4.

- **ARIADNE Experiment (Summer 2023 – Summer 2024):** Creating a collection of custom-made O-ring seals, vacuum check, Helium leak check, radiation shields insertion, flow rate tests, radiation insulation insertion, plasma testing, thermometry sensors connections and gas handling system plumbing for a vertical cryostat and assembling a fully customed horizontal cryostat, both of which are part of the IU efforts in the ARIADNE collaboration to develop the laser-based technique to polarize the ^{3}He gas nuclear spins.

REYES mentorship program, June 2023 – August 2023
A virtual research experience under the supervision of Dr. Sokratis Trifinopoulos from MIT Center for Theoretical Physics. The research project title was "**r-process Nucleosynthesis: Identifying the significant nuclear properties**" and it focused on using a Machine Learning model, NuCLR code, in the prediction of nuclear observables. In particular, the activities included studying phenomena related to Nuclear Astrophysics like the theory of the r-process and describing the nuclear properties that need to be included in the code for the accurate prediction of these processes.

Hong research group, Arizona State University, Spring 2022 – Summer 2023
Calculations of melting temperatures, heat of fusion, heat capacity, thermal expansion, enthalpy, and diffusion via DFT calculations, MD simulations and machine learning, with applications in materials screening and discovery. The work focused on Si-Ge-Sn novel binary and ternary High-entropy alloys with a MS thesis completion

***Quantum Mechanical Engineering lab.*, Arizona State University, Summer 2020 – Summer 2021**
Training in Density Functional Theory (DFT) calculations and Molecular Dynamics (MD) simulations via VASP and LAMMPS codes with a focus on diffusion and melting properties of High Entropy Alloys.

Thin films lab., Physics Department, Jordan University, Jordan, Oct. 2019 – Dec. 2019
Experimental training on thin-films deposition using the flash-evaporation technique and thin-films characterization using X-ray diffraction patterns, scanning electron microscope micrographs, and energy-dispersive spectroscopy.

Awards

OIST Research Internship, 2021
An award to complete a six-month research internship at the Okinawa Institute of Science and Technology (OIST) in Japan during the year 2021.

Al Ghurair Foundation for Education, OLSP Scholarship 2020 – 2023
A full scholarship awarded to study an online MS degree at Arizona State University (ASU)

Training and Certificates

- Machine Learning for Fundamental Physics School (**ML4FP**) 2025. 4 – 8/Aug./2025
- HSF/IRIS-HEP **Software Basics Training** at CERN (Hybrid). 18-20/June/2025
- **National Nuclear Physics Summer School (NNPSS) 2024** at **Indiana University Bloomington**. 15-26/July/2024
- **Summer Lectures 2023 Online Lecture Series on Particle Physics & Condensed Matter Physics**, **University of Tennessee, Knoxville**. 22-23/July/2023
- **The Remote Experience for Young Engineers and Scientists (REYES**). June 26 – August 18/2023
- **Physics Summer Student Seminar Series 2023** at the **Indiana University Center for Exploration of Energy and Matter (CEEM)**. June7- July 19/202
- HPC Online School on **Quantum Computational Materials Science**, MajuLab, Center for Quantum Technology, and Sorbonne University, 16-18/Nov./2020
- **The First Astrophysics Summer School RASSASS I** in the Regional Center for Space Science and Technology Education for Western Asia, Amman, Jordan, 10-12/Sep/2019
- **Third Winter School** in **High Energy Physics** (WISHEPP-III), Collaboration between An-Najah National University, Paris-Sud University, and European Center for Nuclear Physics, 10 – 18/Nov./2018
- **Exoplanets Detection by Transit Method Summer School**, AstroJo Organization, Amman, Jordan, 2-4/May/2018
- **Astronomical Observation and the Basics of Sky Map**, a 40-hour practical training, Jordanian Astronomical Society (JAS), Amman, Jordan, Feb.-May/2017

Languages

Arabic (Native), English (reading, writing, and speaking)

Community - Volunteer

- **Science Students Association, Scientific team, Al-Balqa' Applied University 2016-2019**
- *BAU Science Club Founder: Journal and discussion club that hosts weekly sessions on different science and research topics during the years 2018-2019*
- *Scientific Research event: Event hosting the dean of scientific research, a professor from Biomedical Analysis department and an external guest that focused on the basics of scientific research, paper writing and funding opportunities. During the Second Semester of 2017/2018 School Year*
- *Department of Physics first seminar series: hosting faculty members from the Physics and Chemistry departments during the Second Semester of 2018/2019 School Year*
- **"Mojadedoon" Charity Society 2016-2017**

Professional Memberships

- American Physical Society 2017-2023
- The Optical Society 2019-2020
- Jordanian Astronomical Society 2016-2019

References

- Prof. Jon Urhiem

Professor, Department of Physics, Indiana University Bloomington, United States of America
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- Prof. Michael William Snow

Professor, Department of Physics, Indiana University Bloomington, United States of America
Email: wsnow@indiana.edu

- Dr. Qijun Hong

Assistant professor, School for Engineering of Matter, Transport and Energy, Arizona State University, United States of America
Email: qhong7@asu.edu

- Prof. Asaad Sakhel

Professor, Department of Physics, Al-Balqa' Applied University, Jordan
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